Lab 2 Part 2 report: compressing Qwen3-4B

Student ID: 314561005
Name: 龍偉亮

Setup and size accounting

Model: Qwen/Qwen3-4B, fp16, evaluated only through the provided run_eval.py (WikiText-2 test, seqlen 2048); the PPL code is unchanged. run_eval.py was extended only to call compress.apply_case() before the evaluation, and compression_rate() in evaluate.py (a stub in the handout) now returns the ratio computed by compress.py. Base model: PPL 13.64.

Target layers: the 7 Linear projections (q, k, v, o, gate, up, down) of all 36 blocks, 3.64 B weights, 16 bits each originally. Compression Ratio = 1 - compressed bits / original bits, where the compressed size is counted from the real storage format: integer codes + one fp16 scale per group (+ a zero point per group for asymmetric quantization) for quantized layers; r(m+n) weights (or their quantized size with scales) for SVD layers; removed rows/columns are simply absent for pruned layers. Layers that are not touched cost 16 bits per weight. The model itself keeps fp16 tensors holding the dequantized values (fake quantization), so the numerics are exactly those of the stored format.

Common tool: GPTQ (Hessian-based error compensation) with static groups, symmetric scales, a per-group MSE search over clipping ratios (24 candidates from 1.0 to 0.55), dampening 0.01, block size 128, sequential block-by-block quantization (q/k/v, then o, then gate/up, then down, so each group sees the already-quantized earlier layers). Calibration data: 128 random 2048-token windows of the WikiText-2 **train** split (seed 0); the test split is never touched. The TA confirmed that calibration data is allowed and is not fine-tuning. No weights are trained.

Results

Case	Method	Compression Ratio	PPL	Requirement
1	GPTQ 3-bit, group 128	80.47%	14.55	>75%, PPL<15
2	GPTQ 3-bit, group 512	81.05%	15.79	>=81%, PPL<20
3	whitened SVD + 10-bit factors	51.45% (SVD ratio 24.2%)	23.11	SVD>=20%, overall 50%, PPL<50
4	MLP neuron pruning + GPTQ 3/2-bit	88.07% (pruning 22.4%)	25.23	prune>=20%, overall>=85%, PPL<50

Case 1: GPTQ 3-bit, group 128 (all 7 projections)

Strategy. Every target layer is quantized to 3 bits with a symmetric fp16 scale per 128 weights: 3 + 16/128 = 3.125 bits per weight, i.e. 80.47% compression. Asymmetric 4-bit with groups of 128 costs 4.16 bits and gives only 74.02% (PPL 14.02), which misses the 75% requirement, so 3-bit was the natural choice.

Result. Compression 80.47%, PPL 14.55 (base 13.64). **Analysis.** GPTQ's error compensation plus the clipping search keeps 3-bit nearly lossless (+0.9 PPL). Group size 128 keeps the scale overhead at only 0.125 bit while still isolating outliers.

Case 2: GPTQ 3-bit, group 512

Strategy. The same method with one scale per 512 weights: 3 + 16/512 = 3.031 bits, 81.05% compression (just above the 81% requirement; plain group 128 gives 80.47%). **Result.** 81.05%, PPL 15.79. **Analysis.** Coarser groups lose resolution for outlier-heavy rows (+1.2 PPL versus group 128), but the budget is far from the limit of 20. A 2-bit probe on every layer (group 128) reaches 86.7% but PPL 25.2, so 3-bit with larger groups is the better trade-off at 81%.

Case 3: activation-aware SVD + quantized factors

Strategy. All 252 target layers are replaced by two thin Linear layers W ~ B A. The decomposition is SVD-LLM style: with the input Gram matrix H = X^T X = L L^T collected on the same 128 calibration windows, the SVD is taken on W L, truncated, and mapped back with L^-1, so the truncation minimizes the output error ||(W - W') X||, not the weight error. Rank per layer r = floor(0.78 m n/(m+n)) rounded down to a multiple of 64, which is a 24.2% reduction of the parameter count relative to the original W matrix (the TA's definition of the SVD compression ratio, fp16 factors). The two factors are then quantized to 10 bits (round-to-nearest, symmetric, fp16 scale per 64 values, 10.25 bits per weight), which brings the overall ratio to 51.45%.

Result. SVD ratio 24.2%, overall compression 51.45%, PPL 23.11. **Analysis.** The truncation is the dominant source of error: with fp16 factors and a 21.4% SVD ratio the PPL is already 20.44, while the 10-bit quantization of the factors adds almost nothing. Plain weight-SVD is not usable here; the whitening is what keeps the PPL below 50. The 10-bit width was picked so that the overall ratio just clears 50%.

Case 4: structured MLP pruning + mixed 3/2-bit GPTQ

Strategy. In each block the 9728 MLP intermediate neurons are ranked by rms(activation) x ||down_proj column|| (activations from the calibration set, measured with the already-compressed previous blocks); the 6784 best are kept and the other rows of gate/up and columns of down are physically removed (30.3% of the MLP neurons, 22.4% of all target weights). Structured pruning needs no mask or index, whereas an unstructured mask would cost about 1 bit per weight and defeat the purpose. The remaining weights use GPTQ: attention (q, k, v, o) at 3-bit and MLP at 2-bit, symmetric, group 128.

Result. Pruning rate 22.4%, compression 88.07%, PPL 25.23. A milder prune (7040 neurons, 20.45%) gave 87.81% and 25.74. **Analysis.** The attention projections are cheap to compress (see the sensitivity table) so they get the extra bit; pruning the least important neurons costs little because those neurons are removed together with their 2-bit quantization error. For reference, 2-bit on every layer without pruning gives 25.16 at 86.7%; the pruned model reaches a similar PPL at a higher ratio, although the two differ in several ways (attention is 3-bit here), so I did not isolate the effect of the pruning alone.

Sensitivity of the projection types

Only one projection type quantized to 2-bit (GPTQ, group 128) in all 36 blocks, everything else fp16 (base PPL 13.64):

Projection	q	k	v	o	gate	up	down
PPL	13.71	13.53	14.31	14.04	14.82	15.34	15.18

Overall conclusion

GPTQ-style error-compensated quantization with calibration data is by far the most efficient tool: 3-bit costs less than 1 PPL at 80% compression, and 2-bit gives 87% at PPL 25. SVD is much weaker per bit: even the activation-aware version needs a 24% rank reduction just to reach a PPL of 23 at about 50% overall. Structured pruning is a mild, cheap add-on when the metadata is free. The MLP projections (up, down, gate) are the most sensitive per weight, k/q the least, so a mixed allocation (more bits for the MLP) is preferable. Limitations: compression is simulated (the model keeps fp16 dequantized tensors and the size is accounted from the storage format; no packed kernels); the calibration set is small and from WikiText-2, so results on the WikiText-2 test split may be optimistic relative to other domains; the per-case settings were selected by running a handful of configurations and checking the WikiText-2 PPL, not by an exhaustive search. Peak GPU memory is about 11.6 GB (fp16, fits a 15 GB T4).

Other configurations tried (all with GPTQ, WikiText-2 PPL)

Config	Compression	PPL
4-bit asym, g128	74.02%	14.02
3-bit sym, g128	80.47%	14.55
3-bit sym, g512	81.05%	15.79

Config	Compression	PPL
2-bit sym, g128	86.72%	25.16
SVD 20%, fp16 factors	21.43%	20.44
SVD + 10-bit factors	51.45%	23.11
prune 20.45% + 3/2-bit	87.81%	25.74
prune 22.4% + 3/2-bit	88.07%	25.23